

Dry Eye Syndrome Among Adult patients with and without Diabetes

Ume AO,¹ Odugbo OP,^{2,3} Muhammad RC^{4,5*}

ABSTRACT

Dry Eye Syndrome (DES) undermines quality of life being triggered by multiple factors and is more likely to occur in patients with diabetes than in those without diabetes. The study compared the frequency of Dry Eye Syndrome (DES) among patients with and without diabetes mellitus. One hundred and fifty patients with and without diabetes mellitus were enrolled. Ocular Surface Disease Index (OSDI) scores, tear film break up time (TBUT) test, and Schirmer test were performed on all subjects. Subjects with diabetes had higher relative frequencies of OSDI, TBUT and Schirmer test I. (36.0%, 65.3% and 55.3% respectively Vs 27.3%, 20.7% and 11.3% respectively.) (TBUT: $p < 0.001$; Schirmer test I: $p < 0.001$). There was an association between abnormal OSDI and poor glycaemic control and diabetic retinopathy (OR=0.81, 95% CI: 0.20–0.84, $P = 0.015$; and OR=0.94, 95% CI: 0.47–0.98, $P = 0.044$ respectively); abnormal TBUT and female gender, longer duration of diabetes, poor glycaemic control and diabetic retinopathy (OR=2.18, 95% CI: 1.11–4.71; OR=1.91, 95% CI: 1.25–3.14; OR=3.15, 95% CI: 2.44–5.93; and OR=1.83, 95% CI: 1.09–3.77 respectively); and abnormal Schirmer test I and female gender, longer duration of diabetes, poor glycaemic control and diabetic retinopathy (OR=1.81, 95% CI: 1.16–3.06; OR=2.32, 95% CI: 1.89–5.09; OR=2.42, 95% CI: 1.89–3.70; and OR=1.82, 95% CI: 1.30–3.64 respectively). This study highlights female gender, longer duration of diabetes, poor glycaemic control and retinopathy as predictors for dry eye symptoms among patients with diabetes.

Keywords: Diabetes Mellitus, Dry eye, Schimmer's test

INTRODUCTION

Diabetes mellitus (DM) is a group of metabolic diseases characterized by a patient's inability to utilize carbohydrates due to insufficient or absence of insulin secretion by pancreatic β -cells resulting in elevated blood sugar concentrations.^{1,2} It is one of leading causes of blindness in people aged 20–74 years.³ Dry Eye Syndrome (DES) is one of its most common, but often neglected complications.³

The Tear Film Ocular Surface Dry Eye Work Shop II (TFOS DEWS II) defines dry eye

as “a multifactorial disease of the ocular surface characterized by loss of homeostasis of the tear film accompanied by symptoms in which tear film instability, hyperosmolarity, ocular surface inflammation and damage, play aetiological roles.”⁴ DES undermines quality of life being triggered by multiple factors which include ocular discomfort, visual disturbance and unstable tear film and may be complicated by corneal epithelial defects like erosions or ulcers.⁵

Dry eye syndrome is more likely to occur in patients with diabetes than in those without diabetes.⁶ The reported prevalence of DES among patients with diabetes ranges from 14.4% to 33% using questionnaire-based surveys, while studies involving tests for tear film stability (tear break-up time) and tear volume (Schirmer tests) have generally been shown to have lower prevalence rates.⁷ However, there have been discrepancies in the literature on the reported prevalence of DES among patients with diabetes, due to a lack of

Garki General Hospital,¹ Garki Abuja. Department of Ophthalmology,² College of Health Sciences, University of Jos, Jos, Plateau State, Nigeria. Department of Ophthalmology,³ University of Jos Teaching Hospital, Jos, Plateau State, Nigeria. Department of Ophthalmology,⁴ College of Health Sciences, University of Abuja, Abuja, Nigeria. Department of Ophthalmology,⁵ University of Abuja Teaching Hospital, Gwagwalada, Abuja, Nigeria.

*Corresponding Author: rmchiroma@yahoo.com

Date manuscript was received: 1.6.25

Date manuscript was accepted: 7.1.26

consensus on the appropriate diagnostic criteria and differences in research methodology.⁷

This study aims to compare the relative frequency of dry eye syndrome among patients with and without diabetes, and to determine the associated factors for increased risk of dry eye syndrome among patients with diabetes.

MATERIALS AND METHODS

The study was carried out at the University of Abuja Teaching Hospital, (UATH) Gwagwalada, Abuja, a 350-bed tertiary health care institution located in Gwagwalada, Abuja, Nigeria.

Ethical considerations

Ethical clearance was obtained from the Institutional Health Research Ethics committee and the study adhered to the tenets of the Helsinki declaration. Written informed consent was obtained from all subjects.

Study design and population

This hospital-based comparative clinical study was conducted between February and November 2020. Adults aged 18 years and above with Diabetes Mellitus who visited the Endocrinology outpatient clinic of University of Abuja Teaching Hospital, (UATH) Gwagwalada, Abuja within the study period were recruited and age and sex matched with patients having normal healthy eyes who presented to the outpatient eye clinic within the study period for routine eye checkup.

Inclusion criteria

Adult Patients aged 18 years and older who had been diagnosed with diabetes mellitus (as determined by clinical records of diagnosis using the American Diabetes Association (ADA) Expert Panel diagnostic criteria of DM)⁸ and being treated with oral hypoglycaemic medications, insulin or diet alone, at the endocrinology clinic of UATH, Gwagwalada (for the diabetes group).

Age and sex matched adult patients without diabetes (patients with normal fasting blood sugar (FBS)⁹ and/or random blood sugar (RBS),¹⁰ who had not been previously diagnosed with DM and not on any form of

hypoglycaemic medication) attending the ophthalmology clinic of UATH, Gwagwalada (for the 'no diabetes' group).

Exclusion criteria

The exclusion criteria included patients with conditions that could alter tear secretion and stability such as blepharitis, previous ocular surgeries, thermal and chemical burns, pterygium, trauma, Meibomian and lacrimal gland diseases, allergic and microbial conjunctivitis, conjunctival degenerations and corneal diseases. chronic contact lens wearers, patents with history of smoking, and patients on topical medications such as beta-blockers, topical anaesthetics as well as eye drops containing the preservative benzalkonium chloride, use of systemic medications such as antihistamines, diuretics, anticholinergics phenothiazines, in the six months prior to the study were also excluded from the study. Others are patients with systemic diseases such as Sjögren syndrome, vitamin A deficiency, rheumatoid arthritis, systemic lupus erythematosus and Parkinson's disease.

Sample size

Sample size was determined using the formula for comparison of two independent proportions. Using proportion of patients associated with the outcome of interest = 0.198 (19.8%),¹¹ a desired difference in the outcome of interest of 15.2%, a standard normal deviate for level of significance set at 1.96 (at 95% confidence level) with a possible non-response rate of 10%, A minimum sample size of 300 was estimated with 150 participants in each group.

Study Procedure

A pretested semi-structured interviewer administered questionnaires were used to obtain information on sociodemographic characteristics, type and duration of diabetes, current treatment and possible complications and co-morbidities, current fasting sugar level and recent HbA1c level (done within 3 months).

Ocular Surface Disease Index Score (OSDI):

The subjects were then assessed for dry eye symptoms using ocular surface disease index (OSDI) questionnaire, a valid and reliable 12-item scale designed to provide a rapid assessment of the symptoms of ocular irritation in keeping with dry eye syndrome and their impact on vision-related functioning within the last one week¹²

Assessment of tear break up time (TBUT):

Evaluation of the tear film was carried out in both eyes. This was done with the participant seated at the slit lamp biomicroscope in a dimly lit room and with a relatively constant temperature and humidity (after switching off the fan and air conditioning) so as to minimise reflex tearing.

The pre-corneal tear film was examined for the presence of debris (mucous/oil droplets/debris). Tear break up time (TBUT) measurement was performed without the use of an anaesthetic. A 2% fluorescein strip was wetted with a drop of sterile saline and gently shaken once to remove the excess fluorescein solution, then with the participant looking up, the strip was placed on the temporal one-third of the inferior fornix of the eye for one minute with care taken to avoid contact with the cornea. Subsequently, the participant was asked to blink three times, and then asked to look straight ahead without blinking, the cornea was then examined under low slit lamp magnification ($\times 10$) using a broad beam of cobalt blue filtered light. The time of appearance of the first randomly distributed dry spot (small dark spots) on the cornea from the last blink, was noted using a stop watch, and an average of three timings were recorded as the TBUT.

Schirmer test I:

This was performed five minutes after assessing the TBUT. Before performing this test, excess tears were delicately wiped off the lower cul-de-sac of the eye using a sterile cotton tip applicator. The procedure was then performed (without anaesthesia) by hooking the folded end of a pre-calibrated Schirmer

paper strip (folded at the 5mm end) over the temporal one-third of the inferior fornix of the eye, taking care not to touch the cornea. Thereafter, the participant was asked to gently close both eyes and after 5 minutes, the test strip was removed and the amount of wetting on the strip was recorded.

The adnexae and anterior segments of both eyes were then examined using a pen-torch and diffuse illumination of the slit lamp biomicroscope. The condition of the lids and conjunctival surface were noted. The corneal surface was examined for its sheen. Pupillary reaction was assessed in a dimly lit room using a pen torch and the intraocular pressure was measured using an applanation tonometer.

Examination of the lens and visualisation of the posterior segment were achieved by pupillary dilation with a topical mydriatic (1 – 2 drops/eye) These examinations were performed using the slit lamp and a +78D non-contact fundus lens. If retinopathy was present among the participants with diabetes, it was classified as per ETDRS.

Study definition

The Ocular Surface Disease Index (OSDI)¹³ score was assessed on a scale of 0 to 100 with higher scores representing greater abnormality: Dry eye syndrome (DES) was defined as:

- i. An OSDI score of ≥ 13 ¹³
- ii. TBUT of ≤ 10 sec¹⁴
- iii. Schirmer's test I measurement of ≤ 10 mm/5min¹⁴

A participant was said to have dry eye syndrome if abnormal dry eye parameters were observed in either one or both eyes.

No diabetes was defined as: fasting blood sugar (FBS) of 3.9–6.0mmol/L (70–109mg/dl)⁹ and/or random blood sugar (RBS) of 4.4–7.8mmol/L (80–40mg/dl).¹⁰

Diabetes was defined according to the American Diabetes Association (ADA) Expert Panel⁸ as: fasting blood sugar (FBS) of ≥ 7.0 mmol/L (126mg/dl), random blood sugar (RBS) of ≥ 11.1 mmol/L (200mg/dl) and/or 2-h plasma glucose of ≥ 11.1 mmol/L (200mg/dl) during an oral glucose tolerance test (OGTT).

Poorly controlled diabetes was defined according to the World Health Organization (WHO) as fasting blood sugar (FBS) of ≥ 7.0 mmol/L (126 mg/dl) and/or glycosylated haemoglobin (HbA1c) of $\geq 6.5\%$ (48 mmol/mol).¹⁵

Data management

Collected data were analyzed using Statistical Package for Social Sciences (SPSS) software (SPSS for windows version 25.0; SPSS Inc, Chicago, Illinois). Chi-square test was used for categorical variables. Relative frequencies were calculated for parameters for assessing dry eyes among participants with and without diabetes. The normality of the data was tested using Shapiro-Wilk test. Mann Whitney U-test was applied for comparison between the ocular surface parameters of the independent groups. Spearman rank-order correlation coefficient was performed to determine the association between dry eye parameters and factors such as age, duration of diabetes and level of HbA1c. Multivariate logistic regression was used to predict factors such as age, gender, duration of DM, level of HbA1c and diabetic retinopathy, with dry eye syndrome among participants in the group with diabetes. The level of statistical significance was set at $p < 0.05$ across all parameters.

RESULTS

Three hundred subjects were enrolled in this study, 150 in each group. The mean age was 58.97 ± 11.22 years). The mean ages were 59.08 ± 11.02 and 58.85 ± 11.43 years for those in the group with diabetes and those in the group without diabetes respectively with no significant difference ($\chi^2 = 0.021$, $p = 1.00$). There were more female participants in each group with a male to female ratio of 1:1.72. (Table 1).

A total of 54 participants in the diabetes group had abnormal OSDI scores giving a relative frequency of 36.0% (95% CI: 28.3%–43.7%). Forty-one participants in the group without diabetes had abnormal OSDI scores giving a relative frequency of 27.3% (95% CI: 20.2%–34.4%). ($\chi^2 = 2.603$, $p = 0.107$).

The relative frequency of DES based on TBUT was 65.3% (95% CI: 57.7%–72.9%) among 98 of the 150 participants in the diabetes group, while it was 20.7% (95% CI: 14.2%–27.2%) among 31 of the 150 participants in the group without diabetes. This observed difference was significant ($\chi^2 = 61.05$, $p < 0.001$).

The relative frequency of DES based on Schirmer test I was found to be 55.3% (95% CI: 47.3%–63.3%) in 83 of the 150 participants in the diabetes group and 11.3% (95% CI: 6.2%–16.4%) in 17 of the 150 participants in the group without diabetes, and this observed difference was significant ($\chi^2 = 65.34$, $p < 0.001$). (Table 2)

In the diabetes group, 45 (83.3%), 6 (11.1%) and 3 (5.6%) participants had mild, moderate and severe DES respectively based on the OSDI scores. While in the No diabetes group, 37 (90.2%), 3 (7.3%) and 1 (2.5%) had mild, moderate and severe DES respectively (Figure 1).

Long duration of diabetes was found to be associated with abnormal TBUT (DES) ($r_s = -0.188$, $p = 0.021$) and abnormal Schirmer's test I (DES) ($r_s = -0.174$, $p = 0.034$).

The presence of DR was also shown to be association with abnormal OSDI ($\chi^2 = 6.757$, $p = 0.009$), abnormal TBUT ($\chi^2 = 15.161$, $p < 0.001$) and abnormal Schirmer test I ($\chi^2 = 15.402$, $p < 0.001$).

An association was found between poor glycaemic control and abnormal OSDI score, ($r_s = 0.215$, $p = 0.008$) abnormal TBUT ($r_s = -0.307$, $p < 0.001$) and abnormal Schimmer test ($r_s = -0.239$, $p = 0.003$) (Table 3).

After adjusting for confounders, the likelihood of abnormal TBUT was 2 times higher among female participants (OR=2.18, 95% CI: 1.11–4.71), 1.9 times higher in participants with longer duration of DM (OR=1.91, 95% CI: 1.25–3.14), 3 times higher in participants with abnormal levels of HbA1c (OR=3.15, 95% CI: 2.44–5.93) and 1.8 times higher in participants with diabetic retinopathy (OR=1.83, 95% CI: 1.09–3.77). (Table 4)

The likelihood of having abnormal Schirmer test I measurements was 1.8 times

Table 1: Sociodemographic characteristics of the study population

Variables	Diabetes group Frequency = 150 (100%)	No diabetes group Frequency = 150 (100%)	Total Frequency = 300 (100%)	Test statistic value	p value
Gender					
Male	55 (36.7)	55 (36.7)	110 (36.7)	0.003	1.000
Female	95 (63.3)	95 (63.3)	190 (63.3)		
Total	150 (100.0)	150 (100.0)	300 (100.0)		
Age					
<30	4 (2.7)	4 (2.7)	8 (2.7)	0.021	1.000
30 – 39	3 (2.0)	3 (2.0)	6 (2.0)		
40 – 49	20 (13.3)	20 (13.3)	40 (13.3)		
50 – 59	50 (33.3)	49 (32.7)	99 (33.3)		
60 – 69	45 (30.0)	46 (30.7)	91 (30.3)		
≥70	28 (18.7)	28 (18.7)	56 (18.7)		
Total	150 (100.0)	150 (100.0)	300 (100.0)		
Level of education					
Non-formal	14 (9.3)	15 (10.0)	29 (9.7)	7.570	0.056
Primary	10 (6.7)	16 (10.7)	26 (8.7)		
Secondary	16 (10.7)	29 (19.3)	45 (15)		
Tertiary	110 (73.3)	90 (60.0)	200 (66.7)		
Total	150 (100.0)	150 (100.0)	300 (100.0)		
Occupation					
Skill level 1	17 (11.3)	15 (10.0)	32 (10.7)	0.869	0.833
Skill level 2	24 (16.0)	27 (18.0)	51 (17.0)		
Skill level 3	54 (36.0)	59 (39.3)	113 (37.7)		
Skill level 4	55 (36.7)	49 (32.7)	104 (34.7)		
Total	150 (100.0)	150 (100.0)	300 (100.0)		
Religion					
Christianity	102 (68.0)	109 (72.7)	211 (70.3)	0.783	0.376
Islam	48 (32.0)	41 (27.3)	89 (29.7)		
Total	150 (100.0)	150 (100.0)	300 (100.0)		

†Chi square test, *significant at $p < 0.05$ **Table 2: The relative frequencies of the dry eye parameters among the participants of the groups with and without diabetes**

Variable	Diabetes group n = 150 (100%)	No diabetes group n = 150 (100%)	Total n = 300 (100%)	Test statistic value	†p value
OSDI					
< 13	96 (64.0)	109 (72.7)	205 (68.3)	2.603	0.107
≥ 13	54 (36.0)	41 (27.3)	95 (31.7)		
Total	150 (100.0)	150 (100.0)	300 (100.0)		
TBUT					
≤ 10s	98 (65.3)	31 (20.7)	129 (43.0)	61.05	< 0.001*
> 10s	52 (34.7)	119 (79.3)	171 (57.0)		
Total	150 (100.0)	150 (100.0)	300 (100.0)		
Schirmer's test I					
≤ 10mm/5min	83 (55.3)	17 (11.3)	100 (33.3)	65.34	< 0.001*
> 10mm/5min	67 (44.7)	133 (88.7)	200 (66.7)		
Total	150 (100.0)	150 (100.0)	300 (100.0)		

†Chi square test, *significant at $p < 0.05$

Note: A participant was said to have dry eye syndrome if abnormal dry eye parameters were observed in either one or both eyes;
 Key: OSDI – Ocular Surface Disease Index, TBUT – tear break-up time

higher among the female gender (OR=1.81, 95% CI: 1.16–3.06), 2.3 times higher with longer duration of DM (OR=2.32, 95% CI: 1.89–5.09), 2.4 times higher with poor

DISCUSSION

This study has demonstrated a higher relative frequency of abnormal dry eye parameters (TBUT and Schirmer test I) among patients with diabetes than in those without diabetes attending the endocrinology and eye clinics respectively of UATH, Gwagwalada. The age and gender distributions were similar in both groups with no statistically significant difference establishing a good basis for comparison of the different variables.

The mean age of participants in the group with diabetes was 59.08 ± 11.02 years and this was somewhat similar to studies by Olaniyan *et al.*² and Pooja *et al.*¹⁶ with mean ages of 60.2 ± 10.3 years and 60.2 ± 7.0 years respectively. This may be because majority of the participants (92.7%) had type 2 diabetes which commoner in older individuals.

A reduction in the TBUT is indicative of tear film instability either due to aqueous tear deficiency or Meibomian gland dysfunction. Damage to the lacrimal gland microvasculature accompanied by autonomic neuropathy could impair lacrimation with eventual tear film instability in long standing diabetes.¹⁷ The relative frequency of abnormal TBUT indicating Dry Eye disease was significantly higher among participants with diabetes compared to those without diabetes, and this was similar to some other studies.^{18,19,20 - 24} These findings may be as a result of damage to the corneal and conjunctival epithelia with a subsequent reduction in the number of goblet cells thereby reducing mucin production and the hydrophilic nature of the ocular surface resulting in tear film instability.⁶

Schirmer test values give an estimate of the tear volume and is used to evaluate aqueous tear production. The Schirmer test I, done without anaesthesia estimates total secretion. Studies have shown that intracellular

glycaemic control (OR=2.42, 95% CI: 1.89–3.70) and 1.8 times higher with diabetic retinopathy (OR=1.82, 95% CI: 1.30–3.64). (Table 5).

accumulation of sorbitol leads to cellular oedema and dysfunction with subsequent lacrimal gland damage, dysfunction and a decrease in tear secretion.⁶ Low tear production may also be linked to a dysfunction of the

autonomic nervous system.⁴ This study showed a significant difference in the relative frequency of abnormal Schirmer test I among participants with diabetes compared to those without diabetes ($p < 0.001$) similar to several studies^{18,19,20-24} which therefore suggests an association between inadequate tear volume and diabetes.

Although there was no statistically significant difference in the relative frequencies of abnormal OSDI between both groups, the OSDI scores were significantly higher in the group with diabetes than in the group without diabetes suggestive of a higher severity of dry eye symptoms among patients with diabetes. This was consistent with findings by Divya *et al.*¹⁹ which had mean OSDI scores of 15.1 ± 12.4 and 6.9 ± 6.5 among patients with and without diabetes respectively ($p < 0.001$). Similarly, a study by Manchikanti *et al.*²⁰ had mean OSDI scores of 42.95 ± 17.38 and 16.75 ± 5.45 among patients with and without diabetes respectively ($p < 0.01$). Studies have shown that diabetes is associated with alterations of the corneal and conjunctival epithelia which may result in dry eye symptoms.⁵ This may explain the abnormal OSDI scores seen among participants with diabetes in this study.

Studies have shown significant associations of dry eye syndrome with increasing age and the female gender.^{5,25-27} It has been shown that with increasing age, there is a reduction in lacrimal gland function and this finding was more frequently observed in women than in men.²⁸ The higher prevalence of

dry eye syndrome among females has been attributed to hormonal changes in females associated with menopause.² This study however, showed no significant association between age and the development of dry eye (based on OSDI, TBUT and Schirmer test I)

among participants with and without diabetes similar to what was found in some other study,⁷ there was however a significant association between the female gender and the development of dry eye (based on TBUT and Schirmer test I) among participants in the group

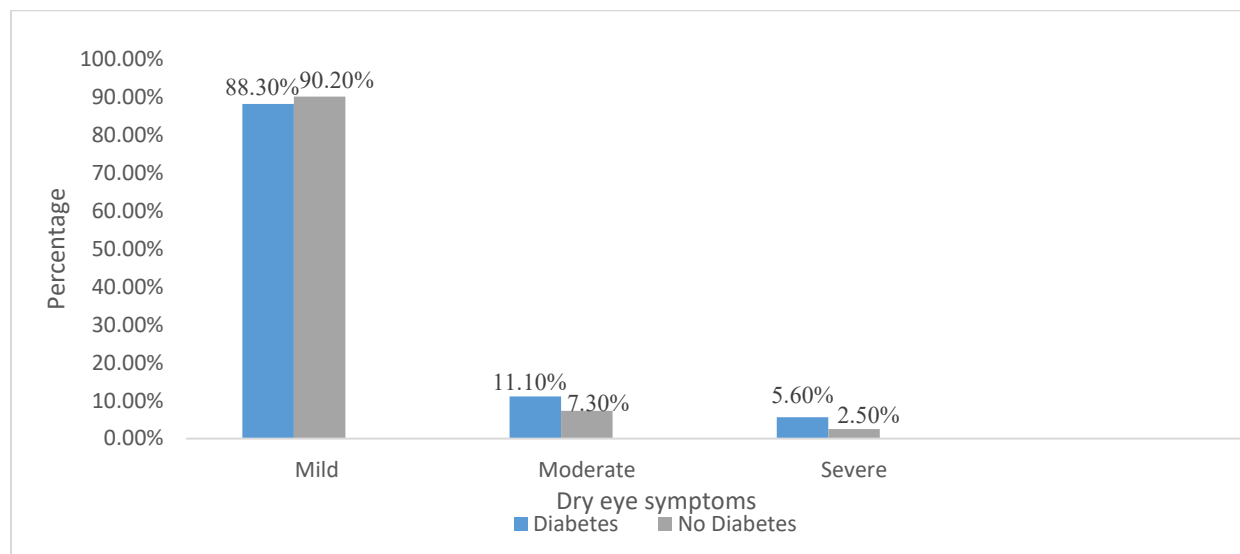


Figure 1: Pattern of distribution of dry eye syndrome according to Ocular Surface Disease Index for the groups with and without diabetes

with diabetes in this study after logistic regression analysis. This finding was similar to a study by Kumar *et al.*²⁷

Dry eye syndrome has been shown to be associated with longer duration of diabetes in several studies.^{3,29,30} This may be due to a higher prevalence of diabetic microvascular complications among these patients which play a role in the pathogenesis of dry eye syndrome.⁵ there were significant associations between the duration of diabetes and abnormal TBUT and Schirmer's test I among the study participants. This was similar to a study by Kamel *et al.*³¹ who found a significant association between the duration of diabetes and TBUT and Schirmer's test ($p=0.03$ and $p=0.002$ respectively).

The glycosylated haemoglobin (HbA1c) is an important measure of long term glycaemic control in diabetes mellitus with the

ability to reveal aggregate glycaemic history in the preceding two to three months.³² It also correlates well with the risk of long term complications such as diabetic retinopathy.³² Studies have shown a significant association between poor glycaemic control and dry eye syndrome.^{4,6} This may be due to microvasculopathy involving the lacrimal gland which impairs lacrimal gland function leading to dry eye syndrome.² In this study, a significant association was observed between all dry eye parameters and poor glycaemic control. Similar findings were observed in some studies.^{4,9} Contrarily, Olaniyan *et al.*² found no association between dry eye syndrome and poor glycaemic control and this was attributed to be probably due to the generally fair glycaemic control among the study participants (mean HbA1c=7.0%) unlike

the poorer glycaemic control observed among participants in this study (mean HbA1c=9.9%). Diabetic retinopathy is a recognized microvascular complication of diabetes, and it has been shown in several studies to be

associated with dry eye syndrome.^{4,6,9} In this study, there was a significant association between dry eye syndrome and diabetic retinopathy. This finding may also be due to microvascular changes involving the lacrimal gland thereby impairing its function.² The

Table 3: Relationship between dry eye parameters and duration of diabetes, glycaemic control and diabetic retinopathy.

Variable	Frequency		Test statistic value	p value
Level of HbA1c [n = 150 (%)]	< 6.5% [n = 63 (%)]	≥ 6.5% [n = 87 (%)]	r _s	p value
OSDI				
< 13	46 (73.0)	50 (57.4)	0.215	0.008*
≥ 13	17 (27.0)	37 (42.6)		
TBUT				
≤ 10s	29 (46.0)	69 (79.3)	-0.307	< 0.001*
> 10s	34 (54.0)	18 (20.7)		
Schirmer's test I				
≤ 10mm/5min	26 (41.2)	57 (65.5)	-0.239	0.003*
> 10mm/5min	37 (58.8)	30 (34.5)		
Duration of DM [n = 150 (%)]	< 10 years [n = 68 (%)]	≥ 10 years [n = 82 (%)]	r _s	p value
OSDI				
< 13	47 (61.1)	49 (59.8)	0.156	0.057
≥ 13	21 (38.9)	33 (40.2)		
TBUT				
≤ 10s	36 (52.9)	62 (75.6)	-0.188	0.021*
> 10s	32 (47.1)	20 (24.4)		
Schirmer's test I				
≤ 10mm/5min	27 (39.7)	56 (68.3)	-0.174	0.034*
> 10mm/5min	41 (60.3)	26 (31.7)		
Diabetic retinopathy [n = 150 (%)]	Present [n = 76 (%)]	Absent [n = 74 (%)]	χ ²	p value
OSDI				
< 13	41 (53.9)	55 (74.3)	6.757	0.009*
≥ 13	35 (46.1)	19 (25.7)		
TBUT				
≤ 10s	61 (80.3)	37 (50.0)	15.161	<0.001*
> 10s	15 (19.7)	37 (50.0)		
Schirmer's test I				
≤ 10mm/5min	54 (71.1)	29 (39.2)	15.402	<0.001*
> 10mm/5min	22 (28.9)	45 (60.8)		

*Significant at p < 0.05

Key: DM – Diabetes Mellitus, HbA1c – glycosylated haemoglobin, OSDI – Ocular Surface Disease Index, TBUT – tear break-up time, r_s – Spearman rank-order correlation coefficient, χ² – Chi square value

Table 4: Multivariate logistic regression analysis of factors predicting abnormal TBUT among participants in the group with diabetes

TBUT	B	p value	OR	95% CI	
Age					
≥ 60 years	0.374	0.450	0.688	0.261	1.816
Ref (< 60 years)					
Gender					
Female	0.822	0.040*	2.175	1.105	4.707
Ref (Male)					
Duration of DM					
≥ 10years	0.807	0.042*	1.910	1.252	3.140
Ref (< 10 years)					
Level of HbA1c					
≥ 6.5%	1.148	0.004*	3.152	2.435	5.926
Ref (< 6.5%)					
DR					
Present	0.912	0.009*	1.834	1.091	3.774
Ref (Absent)					

*Significant at $p < 0.05$

Key: B – Coefficient of the constant, OR – Odds Ratio, CI – 95% confidence interval for the odds ratio, TBUT – tear break up time, DM – Diabetes Mellitus, HbA1c – glycosylated haemoglobin, DR – diabetic retinopathy

Table 5: Multivariate logistic regression analysis predicting abnormal Schirmer's test I among participants in the group with diabetes

Schirmer's test I	B	p value	OR	95% CI	
Age					
≥ 60 years	0.190	0.322	0.612	0.232	1.617
Ref (< 60 years)					
Gender					
Female	0.912	0.014*	1.810	1.159	3.064
Ref (Male)					
Duration of DM					
≥ 10years	0.843	0.036*	2.321	1.885	5.087
Ref (< 10 years)					
Level of HbA1c					
≥ 6.5%	0.885	0.019*	2.422	1.887	3.696
Ref (< 6.5%)					
DR					
Present	0.897	0.015*	1.815	1.303	3.636
Ref (Absent)					

*Significant at $p < 0.05$

Key: B – Coefficient of the constant, OR – Odds Ratio, CI – 95% confidence interval for the odds ratio, DM – Diabetes Mellitus, HbA1c – glycosylated haemoglobin, DR – diabetic retinopathy

hospital-based nature of the study and non-probability sampling technique are considered limitations of this study.

Conclusion

The study has shown that dry eye syndrome is commoner among patients with diabetes compared to general population. Given the impact of tear film instability on

quality of vision, corneal integrity and postoperative outcomes, ophthalmologists should routinely assess and address dry eyes in diabetic patients. Incorporating dry eye evaluation into both medical and surgical management plans of diabetic patients is essential to optimize treatment outcomes. Management of dry eye should therefore be considered an integral component of diabetic eye care.

REFERENCES

1. American Diabetes Association. Diagnosis and Classification of Diabetes Mellitus. *Diabetes Care* 2009; 39 (Suppl 1): S62 – S7.
2. Olaniyan S, Fasina O, Bekibele C, Ogundipe A. Relationship between dry eye and glycosylated haemoglobin among diabetics in Ibadan, Nigeria. *Pan Afr Med J* 2019; 33:14.
3. Manaviat M, Rashidi M, Afkhami-Ardekani M, Shoja M. Prevalence of dry eye syndrome and diabetic retinopathy in type 2 diabetic patients. *BMC Ophthalmol* 2008; 8:10–3.
4. Craig JP, Nichols KK, Akpek EK, Caffery B, Dua HS, Joo C, et al. TFOS DEWS II Definition and Classification. *Ocul Surf* 2017; 15:276–283.
5. Zhang X, Zhao L, Deng S, Sun X, Wang N. Dry Eye Syndrome in Patients with Diabetes Mellitus: Prevalence, Etiology, and Clinical Characteristics. *J Ophthalmol* 2016; 2016:8201053.
6. Yazdani-Ibn-Taz MK, Han MM, Jonuscheit S, Collier A, Nally JE, Hagan S. Patient-reported severity of dry eye and quality of life in diabetes. *Clin Ophthalmol* 2019; 2019:217–224.
7. Ribeiro MV, Barbosa FT, Ribeiro LE, Lacet CM, Lyra JM, Ribeiro EA. Clinical features of diabetic patients with dry eye disease in a community in Maceio: a cross – sectional study. *Rev Bras Oftalmol* 2016; 75:121–126.
8. American Diabetes Association. Diagnosis and classification of diabetes mellitus. *Diabetes Care* 2008; 31: S55-S60.
9. American Diabetes Association. The American Diabetes Association and World Health Organization Classifications for Diabetes. *Diabetes Care* 2002; 25:951-955.
10. Vieira G. What are normal blood sugar levels? (Internet). 2018 (accessed 2019 Dec 3). Available from <https://diabetesstrong.com/what-are-normal-blood-sugar-levels>
11. Li H, Pang G, Xu Z. (Tear film function of patients with type 2 diabetes). *Zhongguo Yi Xue Ke Xue Yuan Xue Bao* 2004; 26:682–686. (Chinese).
12. Walt J, Rowe M, Stern K. Evaluating the Functional Impact of Dry Eye: The Ocular Surface Disease Index. *Drug Inf J* 1997; 31:1436.
13. Miller K, Walt J, Mink D, Satram-Hoang S, Wilson S, Perry H, et al. Minimal clinically important difference for the ocular surface disease index. *Arch Ophthalmol* 2010; 128:94–101.
14. Kanski J. Dry eye. In: Bowling B, ed. *Clinical Ophthalmology: A systematic approach* 8th ed. London, UK: Elsevier 2016; p. 124.
15. World Health Organization. Classification of diabetes mellitus 2019 (Internet). 2019 (accessed 2019 Aug 21). Available from <http://who.int/publications-detail/classification-of-diabetes-mellitus>
16. Pooja HV, Subrahmanya BK, Gowda HT. Correlation of dry eye status with the duration of diabetes and the severity of diabetic retinopathy in type 2 diabetes mellitus patients. *Indian J Clin Exp Ophthalmol* 2019; 5:12–5.
17. Najafi L, Malek M, Valojerdi A, Aghili R, Khamseh M, Fallah A, et al. Dry eye and its correlation to diabetes microvascular complications in people

- with type 2 diabetes mellitus. *J Diabetes Complications* 2013; 27:459–462.
18. Aljarousha M, Badarudin NE, Che Azemin MZ. Comparison of Dry Eye Parameters between Diabetics and Non-Diabetics in District of Kuantan, Pahang. *Malays J Med Sci* 2016; 23:72–77.
 19. Divya K, Dhivya N, Ganesh MR, Sundar D. Dry eye disease and tear dysfunction in patients with type 2 diabetes: A hospital-based study from South India. *Sudanese J Ophthalmol* 2019; 11:14–18.
 20. Manchikanti V, Kasturi N, Rajappa M, Gochhait D. Ocular surface disorder among adult patients with type II diabetes mellitus and its correlation with tear film makers: A pilot study. *Taiwan J Ophthalmol Epub ahead of print* (accessed 2020 Dec 15). DOI: 10.4103/tjo.tjo_56_20
 21. Kesarwani D, Rizvi SW, Khan AA, Amitava AK, Vasenwals SM, Siddiqui Z. Tear film and ocular surface dysfunction in diabetes mellitus in an Indian population. *Indian J Ophthalmol* 2017; 65:301–304.
 22. Zhang M, Xiang Y. Analysis of ocular surface dysfunction in patients with type 2 diabetes mellitus. *Int Eye Sci* 2020; 20:1853–1857.
 23. Yoon KC, Im SK, Seo MS. Changes of tear film and ocular surface in diabetes mellitus. *Korean J Ophthalmol* 2004; 18:168–174.
 24. Raman P, Jasuja M, Singal AK. Dry Eye Occurrence in Type II Diabetics and Non-diabetics: A Prospective Cohort Study. *Acta Scientific Ophthalmology* 2021; 4:30–35.
 25. Paulsen AJ, Cruickshanks KJ, Fischer ME, Huang GH, Klein BE, Klein R, et al. Dry eye in the beaver dam offspring study: prevalence, risk factors, and health-related quality of life. *Am J Ophthalmol*. 2014; 157:799–806.
 26. Schaumberg DA, Dana R, Buring JE, Sullivan DA. Prevalence of dry eye disease among US men: estimates from the Physicians' Health Studies. *Arch Ophthalmol* 2009; 127:763–768.
 27. Kumar A, Vyas M. Dry eye and Corneal Sensitivity in Patients with Type 2 Diabetes Mellitus. *J Med Sci Clin Res* 2018; 6:32383–32387.
 28. Paiva CS. Effects of Aging in Dry Eye. *Int Ophthalmol Clin* 2017; 57:47–64.
 29. Khetawani D, Karambelkar VH, Gadre GA. Ocular Surface Disorders in Type 2 Diabetes Mellitus. *Int J Contemp Med Res* 2019; 6:C9–C12.
 30. Nora B. The Incidence of Dry Eye Disease Related to Long Term Diabetes Mellitus Tip 2. *JOJ Case Stud* 2017; 1:555571.
 31. Kamel SS, Mohammed TH, El Zankalony YA, Saad AH. Prevalence of dry eye in diabetics. *J Egypt Ophthalmol Soc* 2017; 110:77–82.
 32. Sherwani SI, Khan HA, Ekhzaimy A, Masood A, Sakharkar MK. Significance of HbA1c test in diagnosis and prognosis of diabetic patients. *Biomark Insights* 2016; 11:95–104.